

TrES-3b

R-0510

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_240716 · created 2026-07-20T19:54:15+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460507.8543 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.088 +/- 0.056
Transit depth (Rp/Rs)^2	0.77 +/- 0.98 [%]
Orbital Inclination (inc)	79.52 +/- 2.92
Airmass coefficient 1 (a1)	1.25 +/- 0.33
Airmass coefficient 2 (a2)	-0.18 +/- 0.23
Scatter in the residuals of the lightcurve fit is	25.44 %
Best Comparison Star	#1 - [442, 421]
Optimal Aperture	2.91
Optimal Annulus	10.96
Transit Duration (day)	0.02 +/- 0.025

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.088 $\pm$ 0.056 vs 0.16309 (deviation 1.34 σ)
Duration fit vs published	0.02 d vs 0.0567 d (deviation 1.47 σ)
Mid-time O-C	1.4 min
Depth SNR	1.6
Residual scatter	25.44 %

Light curve

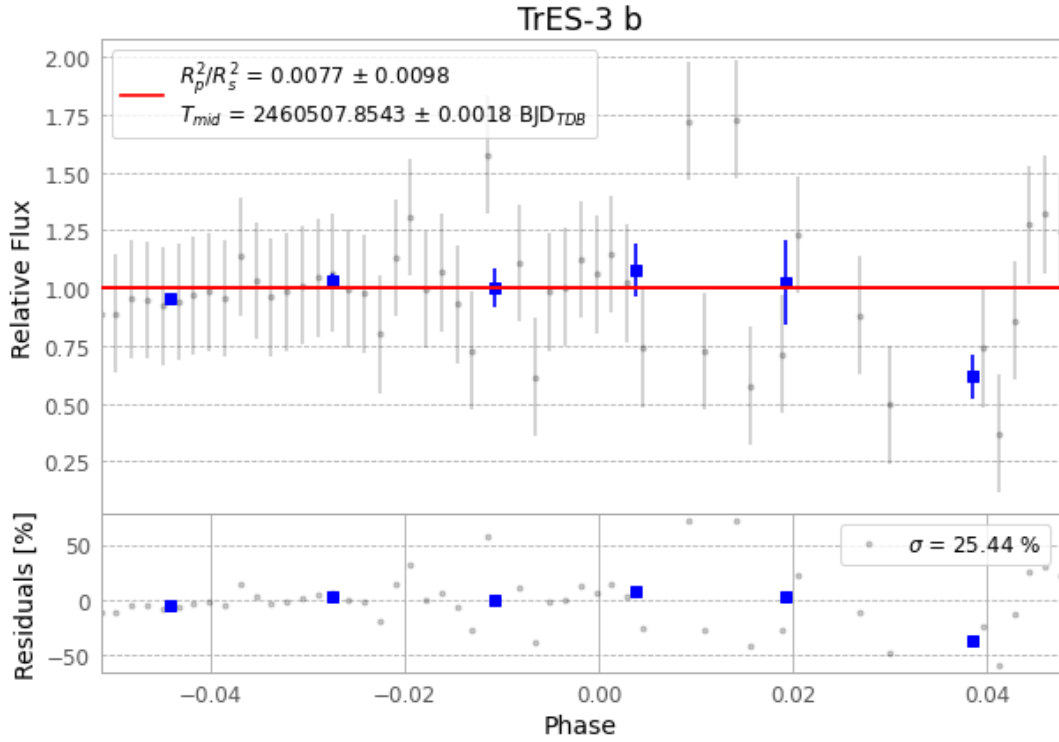


Figure 1 — FinalLightCurve_TrES-3 b_2024-07-15.png (SHA-256 68c5f7d4ed8a1640...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [334, 264], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 0fe9f474641fd585...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

68 FITS frames (45 MB), TRES-3240716064815.FITS → TRES-3240716100916.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-240716.log (fe1880b67b86...), FinalLightCurve_TrES-3 b_2024-07-15.png (68c5f7d4ed8a...), FinalLightCurve_TrES-3 b_2024-07-15.csv (b7f41b4cac1c...)

Compute resources

Wall time 155.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-20 19:54Z.